



Improving Low-Light Face Recognition using DeepFace Embedding and Multi-Layer Perceptron

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Abstract

Facial recognition systems often struggle under extreme lighting conditions, which distort facial features and reduce recognition accuracy. This study introduces a novel integration of DeepFace embeddings with a lightweight Multi-Layer Perceptron (MLP) classifier tailored to improve facial recognition under extreme lighting conditions. This combination has not been explored in previous studies and offers a compact alternative to conventional CNN-based methods. The Labeled Faces in the Wild (LFW) dataset was augmented using rotation, flipping, and lighting variations, and further enhanced with CLAHE for improved contrast under poor illumination. The resulting 128-dimensional DeepFace embeddings were classified using a four-layer MLP with LeakyReLU activation, Batch Normalization, and Dropout. The model was evaluated across three data-splitting schemes (70:30, 80:20, and 90:10), with the 80:20 configuration achieving the highest accuracy of 95.16%. Compared to the baseline CNN, the proposed method demonstrated superior robustness to illumination variations. This makes the proposed model suitable for real-time applications such as biometric authentication and AI-based surveillance systems.

Keywords: DeepFace; face recognition; low-light; MLP; neural network

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1. Introduction

Face recognition is one of the most critical technologies in artificial intelligence-based systems, especially in security, such as automatic surveillance and access control [1]. In addition, this technology is also widely used in biometric authentication and automatic attendance systems based on facial images [2]. Although various methods have successfully improved face detection accuracy, the main challenge remains in uneven lighting conditions. Lighting variations can cause facial feature distortion and loss of important visual information, significantly reducing system performance [3]. Previous studies have proposed methods such as Transfer Learning with CNN architecture, a Classifier, and DeepFace. Lidya et al [4] used DenseNet169 via transfer learning to detect face masks and achieved high accuracy. However, since the dataset used had uniform lighting conditions, this study did not evaluate the model's robustness against visual distortions caused by extreme lighting conditions commonly found in real-world scenarios, such as

shadows or backlighting. Lalitha et al [5] used Haar Cascade for a real-time facial attendance system. Although computationally efficient, this approach heavily relies on edge features, which are easily disrupted by noise and low lighting, leading to detection failures in uncontrolled environments like dimly lit rooms or at night. Abdulatif et al. [6] evaluated the MobileNetV2 and InceptionV3 CNNs for face recognition and achieved high accuracy. However, these models showed a performance decline when tested on low-light images due to the lack of specialized preprocessing, and they have high computational requirements that limit their application in embedded or real-time systems. Badreddine et al. [7] applied MLP with DCT-based input and achieved high accuracy on a limited dataset. However, this model does not utilize modern feature representations such as face embeddings and still relies on classical classification approaches that are less adaptive to visual distortions caused by extreme lighting conditions. The evaluation was also limited to homogeneous data, thus failing to demonstrate robustness in real-world environments.

Madhavi et al [8] applied DeepFace in a facial expression recognition system, where testing was performed using 200 facial images captured directly from a 3-minute webcam video. These images were collected during the live operation of the system, allowing performance evaluation under actual usage conditions. While DeepFace performs well in expression recognition, its resilience to visual noise and uneven lighting has not yet been experimentally proven[9]. Embedding-based models like DeepFace are widely used in facial recognition because they can generate concise and efficient feature representations. The DeepFace model employed in this study generates a 128-dimensional face embedding vector, which effectively represents discriminative facial features in a compact and standardized format. This embedding dimension was chosen due to its balance between computational efficiency and high recognition performance across varying lighting conditions. However, the performance of this method is highly influenced by the visual quality of the input images. Without adequate preprocessing, such as contrast enhancement or lighting adjustment[10], the extracted features tend to be disrupted by shadows or noise, thereby reducing accuracy under extreme lighting conditions [11]. Other approaches have attempted to improve class separation using margin-based loss function modifications. While effective in large-scale classification scenarios, these approaches remain dependent on a large amount of data and do not directly address the issue of extreme lighting conditions from a visual perspective [12]. In contrast to previous works that rely on complex CNN architectures or omit illumination-aware preprocessing, this study introduces a hybrid face recognition pipeline that integrates DeepFace embeddings with a lightweight Multi-Layer Perceptron (MLP) classifier [13]. This combination offers better computational efficiency and robust performance under extreme lighting conditions. The novelty of this research lies in combining modern embedding techniques with CLAHE-based contrast enhancement [14] and an efficient MLP architecture

specifically optimized for low-light facial recognition. To our knowledge, such an integration has not been explicitly explored in previous studies focusing on illumination challenges in face recognition. Evaluation is conducted on the LFW dataset with three data-splitting scenarios to measure the model's robustness in lighting variations. In recent years, significant progress has been made in addressing the challenges of low-light face recognition. Transformer-based architectures, such as Retinexformer [15], integrate self-attention mechanisms with retinex decomposition to enhance feature clarity under poor illumination conditions. Meanwhile, diffusion models like Diff-Retinex [16] leverage a generative technique to reconstruct fine-grained facial details lost in low-light images. Lightweight networks, such as Edgeface [17], combine efficiency and accuracy, enabling deployment on edge devices with unlimited computational resources. Additionally, using margin-based classification, such as ArcFace [18] and its variants, has improved robustness to intra-class variations caused by lighting inconsistencies. These innovations collectively illustrate a shift towards hybrid methods that fuse preprocessing, efficient embeddings, and adaptive loss functions to enhance face recognition in real-world low-light scenarios. To our knowledge, this is the first study that integrates DeepFace embeddings, CLAHE preprocessing, and a lightweight MLP classifier to address the specific challenge of facial recognition under low-light conditions.

2. Methods

This study adhered to the workflow of the Machine Learning Life Cycle (MLLC) [19], which includes data collection, pre-processing, feature extraction using DeepFace, model training with the Multi-Layer Perceptron (MLP) trained with the backpropagation algorithm [20], and model performance evaluation. The research methodology workflow is presented in Figure 1.

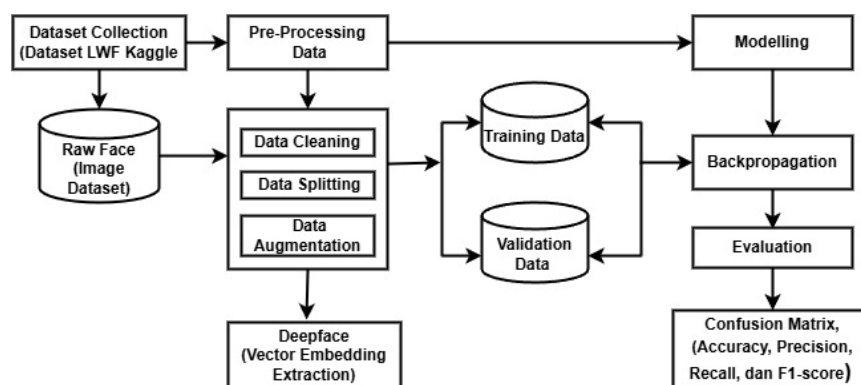


Figure 1. Research Workflow

Based on Figure 1, the workflow follows the standard Machine Learning Life Cycle, from data acquisition to model deployment. Each stage included structured operations such as augmentation, CLAHE

preprocessing, DeepFace embedding with VGG-Face backbone, and classification using a four-layer MLP with LeakyReLU activations and Adam optimizer for convergence stability.

2.1 Data Collection

The face dataset used in this study is the Labeled Faces in the Wild (LFW) People Dataset obtained from Kaggle [21], with a total of 1,680 initial images covering various lighting conditions. The LFW dataset was chosen because it is a widely used benchmark in facial recognition research and contains diverse samples collected from real-world scenarios, such as news photographs and spontaneous captures, rather than controlled laboratory settings. This diversity ensures that the dataset represents different facial poses, backgrounds, and illumination levels.

To strengthen the robustness of the proposed model, images with varied lighting conditions were specifically emphasized. For instance, several images in LFW exhibit strong shadows, backlighting, or uneven illumination, which resemble real surveillance environments. By incorporating such data, the model is

expected to recognize faces more accurately under challenging conditions. Furthermore, the dataset consists of 28 identity classes, with an average of 60 images per class, which allows sufficient intra-class variation for effective feature learning. Although the dataset size is relatively small compared to million-scale facial datasets, its heterogeneity provides an appropriate balance between complexity and feasibility for model training in this study

2.2 Data Pre-processing

To increase data diversity and improve model generalization, this study applied data augmentation techniques including random horizontal flipping ($p = 0.5$), rotation ($\pm 15^\circ$), and brightness variation ($\pm 20\%$) [22]. Examples of augmented facial images are presented in Figure 2.



Figure 2. Example of augmented facial images using flip, rotation, and brightness variation, which simulate diverse lighting and pose conditions to enhance robustness

These transformations simulate facial orientation and lighting variations, allowing the model to learn robust features and reduce overfitting without requiring manual dataset expansion. After the augmentation process, the dataset was divided into 28 identity classes with an average of 60 images per class under different visual conditions [23]. Augmentation generates diverse image conditions that help the model recognize faces across lighting and pose scenarios. In addition to augmentation, CLAHE (Contrast Limited Adaptive Histogram Equalization) was applied to enhance local contrast, particularly in dark facial areas. The method was configured with a clip limit of 2.0 and a grid size of 8×8 to avoid over-amplifying noise [24]. This enhancement improves visibility in shadowed regions and ensures facial features remain distinguishable under low-light conditions. To enhance the quality of facial images before feature extraction, preprocessing was applied using the CLAHE (Contrast Limited Adaptive Histogram Equalization) technique. This method improves local contrast and reduces the impact of uneven lighting, making facial details more distinguishable for the model. The improvement

achieved through CLAHE preprocessing is presented in Figure 3.



Figure 3. CLAHE enhancement improves local contrast in low-light facial images for better feature extraction

CLAHE adjusts the brightness distribution by equalizing histograms in smaller image regions, resulting in clearer and more balanced visual output. This process enhances important facial features while minimizing noise amplification. As a result, the model receives inputs with more consistent contrast, contributing to better recognition performance in

subsequent stages. After preprocessing improves the quality of facial images, the next step is to compare the proposed approach with an established benchmark. For this purpose, the DeepFace model is adopted as a baseline architecture in the evaluation. As a baseline for comparison, the DeepFace architecture was employed in this study. This model has been widely applied in face recognition tasks due to its strong feature extraction capability and proven robustness across various datasets. The architecture of the DeepFace model is presented in Figure 4.

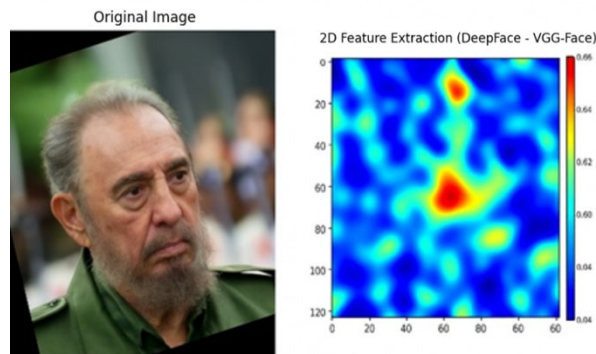


Figure 4. Architecture of the DeepFace model used to generate 128-dimensional facial embeddings, which encode discriminative facial features in a compact vector format

The 128-dimensional feature vectors generated by DeepFace ensure compatibility with lightweight classifiers such as the Multi-Layer Perceptron (MLP). These vectors are forwarded to the MLP for identity classification based on feature similarity. The system can generalize across diverse facial inputs by separating feature extraction from classification while maintaining high recognition accuracy.

2.3. Model Development

The face detection model in this study was designed to address accuracy challenges in extreme lighting conditions. A combination of DeepFace as a feature extractor and Multi-Layer Perceptron (MLP) as a classifier [25] was used. DeepFace was chosen for its ability to generate efficient and standardized fixed-dimensional face embeddings, facilitating modular classification processes. MLP was selected due to its computationally lightweight architecture, stability during training on small datasets, and flexibility in combining with embedding vectors without requiring additional convolutional layers. This approach suits real-time implementations and systems with limited resources [26]. The network structure used consists of four main layers: a layer with 256 neurons, followed by 128 neurons, then 64 neurons, and finally an output layer with 28 neurons corresponding to the number of individual classes in the dataset. Each layer is accompanied by batch normalization techniques to improve model performance during training and stabilize data distribution. Dropouts reduce the risk of overfitting by randomly deactivating some neurons during training. The activation function used is LeakyReLU [27], which allows negative values to flow

to the next layer and helps prevent the vanishing gradient problem [28]. The training process consisted of 40 epochs, each with a batch size 32. The learning rate was set to 0.001 [29] and adjusted to ensure that weight updates remained stable and converged efficiently. The Adam (Adaptive Moment Estimation) algorithm [30] was used to optimize the training process, which adjusts weight updates based on the first and second moments of the gradient, making training more stable against gradient variations between iterations. All training was performed using face embedding data extracted from DeepFace, which had previously been processed using the CLAHE contrast enhancement method to improve lighting quality (detailed explanation in the preprocessing section). The augmentation strategy in the preprocessing stage is crucial for addressing the limited data and improving the model's robustness against extreme lighting variations commonly found in real-world environments. The training process begins with the forward pass stage, which calculates the activation input for each neuron based on the weights and biases from the previous layer. The calculation of the input value (pre-activation) at neuron i in layer l is expressed in Formula 1.

$$z_i^{(j)} = \sqrt{\sum_j w_{ij}^{(j)} a_{\frac{(\ell-1)}{j}}^{(j)} + b_i^{(j)}} \quad (1)$$

The value $z_i^{(j)}$ is the input before activation, which is then passed to the LeakyReLU activation function to produce the neuron output $a_{\frac{(\ell)}{i}}^{(j)} = f(z_i^{(j)})$. The activation value becomes the input for the next layer, and in the final layer, it will be used as the prediction output y_i is compared with the target label in the evaluation stage. After obtaining the output from the forward pass stage, the next step is to evaluate how far the model's prediction differs from the actual target. This evaluation uses the Mean Squared Error (MSE) loss function, which is expressed in Formula 2.

$$\mathcal{L} = \frac{1}{2} \sum_i (y_i - \hat{y}_i)^2 \quad (2)$$

This loss function measures the global error of the model against all outputs. The value of $\mathcal{L}$ will be used as the basis for calculating the gradient during the backpropagation process. The following process is backpropagation, which estimates the error value at each neuron. The error at neuron i in layer l is calculated based on the derivative of the loss function with respect to the output, multiplied by the derivative of the activation function, as presented in Formula 3.

$$\delta \frac{(j)}{j} = \frac{\delta \mathcal{L}}{\delta a_{\frac{(j)}{i}}} \cdot \hat{f}'(z_i^{(j)}) \quad (3)$$

The value δ indicates how much the neuron contributes to the total error. This information is used to update the model parameters. The final step is weight updating, which is performed iteratively to adjust the model to minimize error. Weights are updated based on the learning rate, η , error value δ , and the output of the

neurons in the previous layer, as presented in Formula 4.

$$z_i^{(j)} = \sqrt{\sum_j w_{ij}^{(j)} a_{(l-1)}^{(j)} + b_i^{(j)}} \quad (4)$$

This calculation process is repeated for 40 epochs using a batch size of 32 and the Adam algorithm for optimization, with adaptive weight adjustment based on the gradient moment. Thus, the model gradually minimizes error and improves prediction accuracy.

2.4. Evaluation

Model testing in this study was conducted using test data obtained from three data-splitting scenarios, namely 70:30, 80:20, and 90:10. The purpose of this scheme is to evaluate model performance objectively and avoid using training data in the testing process. Thus, the evaluation results reflect the model's ability to generalize new data that has never been seen before. This study did not implement cross-validation schemes such as k-fold or stratified k-fold due to several practical constraints. First, although the dataset contains 1,680 images, it is partitioned into 28 individual identity classes, resulting in a relatively small number of samples per class. Applying k-fold cross-validation under these conditions may introduce high variance and inconsistent fold distributions, particularly when class identities overlap. Second, the computational cost associated with repeatedly training the MLP model across multiple folds becomes substantial when combined with data augmentation and CLAHE preprocessing. Therefore, this study opted for three distinct data split scenarios (70:30, 80:20, and 90:10) to simulate varying training-to-testing ratios efficiently.

Future work employing larger and more diverse datasets is encouraged to apply stratified k-fold cross-validation for improved statistical robustness and generalization. Several evaluation metrics were used to assess the model's performance, namely accuracy, precision, and recall [31]–[33]. Accuracy measures the proportion of correct predictions compared to the total data used. Precision indicates the model's accuracy in classifying positive data, which is the ratio of correct positive predictions to all positive predictions. Recall measures the model's ability to detect all positive data in the dataset. This evaluation is essential to determine whether the developed model works well or needs improvement to achieve more optimal results. A conventional CNN baseline model consisting of three convolutional layers and two fully connected layers was employed for comparison. This baseline model was trained directly on facial images without using DeepFace embedding. The evaluation results show that the proposed DeepFace with an MLP classification model demonstrates more stable and accurate performance, especially under extreme lighting conditions. This demonstrates the effectiveness of combining facial embedding and a lightweight architecture, such as an MLP, in improving the

robustness of facial recognition systems against lighting variations.

3. Results and Discussion

3.1 Model Implementation

This study implemented a Multi-Layer Perceptron (MLP) artificial neural network architecture using the Backpropagation training method. The implementation results of the MLP architecture are presented in Figure 5.

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 256)	33024
batch_normalization (Batch Normalization)	(None, 256)	1024
dropout_4 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 128)	32896
batch_normalization_1 (Batch Normalization)	(None, 128)	512
dropout_5 (Dropout)	(None, 128)	0
dense_5 (Dense)	(None, 64)	8256
dropout_6 (Dropout)	(None, 64)	0
dense_6 (Dense)	(None, 28)	1820

Figure 5. Multi-Layer Perceptron (MLP) architecture consisting of four hidden layers used for identity classification based on DeepFace embeddings

The model structure consists of four hidden layers, each with 256, 128, 64, and 28 neurons. Each layer is equipped with Batch Normalization and Dropout techniques to improve training stability and reduce the risk of overfitting. The activation function used is LeakyReLU, which effectively overcomes the vanishing gradient problem. The model input is a 1,680-dimensional vector resulting from feature extraction from facial images. Overall, the model has 77,532 trainable parameters. This architecture provides the foundation for evaluating how well the network can capture patterns from the training data. The training process was monitored using accuracy and loss as key performance indicators to assess its effectiveness. Accuracy represents the proportion of correctly classified samples, while loss quantifies the prediction errors produced during training. Observing the progression of both metrics across epochs provides valuable insights into the model's convergence behavior and its ability to generalize effectively to unseen data. The accuracy and loss trends observed during 40 epochs of training are presented in Figure 6.

The training graph demonstrated rapid learning within the first 10 epochs. Accuracy surpassed 90%, while loss approached convergence. Adam optimizer and a batch size of 32 contributed to stable training with minimal overfitting, as indicated by synchronized loss behavior between training and validation. The training and validation accuracy values increased gradually and stabilized at high levels, showing that the model

successfully captured data patterns without difficulty. The loss values for both training and validation decreased simultaneously, indicating smaller prediction errors as training progressed. Moreover, there is no indication of overfitting since the validation loss did not

rise after reaching its minimum. These results confirm that Batch Normalization and Dropout improved generalization, enabling the model to perform well on unseen data under different lighting conditions.

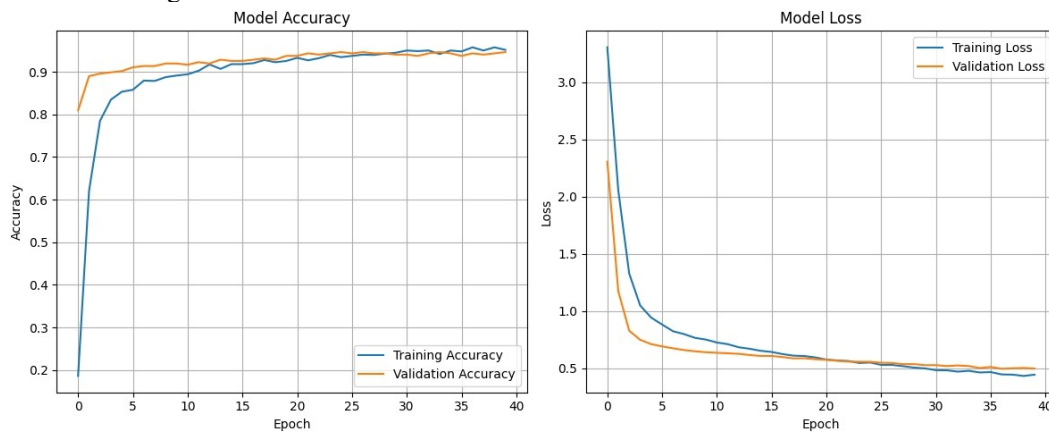


Figure 6. Accuracy and loss trends during 40 epochs of training, showing stable learning progression and convergence behavior for both training and validation sets

3.2 Model Performance Evaluation

Model performance evaluation was conducted by dividing the dataset into three scenarios: 70:30, 80:20, and 90:10 for training and testing data. The evaluation results are presented in Table 1.

Table 1. Model Performance Evaluation Result

Data Split	Accuracy	Precision	Recall
70:30	93.01	93.01	93.01
80:20	95.16	95.16	95.16
90:10	90.40	90.40	90.40

Table 1 shows the model performance results for three data-splitting scenarios: 70:30, 80:20, and 90:10. The evaluation was conducted using accuracy, precision, and recall metrics. The 80:20 data splitting scheme yielded the best results with accuracy, precision, and recall of 95.16%. Statistically, this indicates that this scheme's training and testing data amounts are balanced. In the 70:30 scheme, the test data amount is greater, making model evaluation more stable. However, because the training data is smaller, the model does not learn enough from data variations, decreasing to 93.01%. Meanwhile, in the 90:10 scheme, the training data is indeed the largest, but the insufficient test data makes the evaluation results less accurate and does not represent the model's actual performance. This is evident from the decrease in performance to 90.40%. The 80:20 scheme provided the most effective balance among the three tested schemes. The model has enough data to learn well, and the evaluation results are reliable because the testing data is sufficient. In other words, this scheme allows the training and testing processes to run optimally, resulting in superior performance compared to other schemes. Therefore, the 80:20 scheme is considered the most appropriate because the training data is sufficient for the model to learn effectively, and the testing data is also enough to accurately assess the model's performance.

3.3 Confusion Matrix Analysis

To further assess the classification performance of the proposed model, a confusion matrix was constructed based on the training and testing data split. This visualization provides a concise overview of how the system distinguishes between 28 individual classes and serves as an initial reference point before analyzing specific classification strengths and weaknesses. The confusion matrix for the 80:20 split is presented in Figure 7.

The confusion matrix details the model's ability to classify 28 individual classes. Each row corresponds to the actual label, while each column represents the predicted label. Most predictions align along the main diagonal, showing that the model correctly identified the majority of samples. However, certain classes remain prone to misclassification. For example, the Tom Ridge class is often mistaken for Tom Daschle, likely due to overlapping visual characteristics such as lighting or facial angle. Similarly, George W. Bush was occasionally classified as Gerhard Schroeder, and Tommy Thompson's images were sometimes misidentified as other individuals with comparable features. Despite these errors, the model demonstrates strong overall performance, indicating its practicality for real-world scenarios. Further improvements in training data and augmentation techniques could enhance its robustness even more.

3.4 Comparison with Previous Studies

After evaluating the model's performance across various data-splitting scenarios, the next step is to conduct a comparative analysis with several previous studies. This comparison aims to assess the effectiveness of the proposed approach's combination of DeepFace and Multi-Layer Perceptron (MLP) trained with backpropagation in face detection under varying lighting conditions. Previous studies have

employed various methods and types of datasets, each offering distinct approaches in terms of algorithm complexity, training data volume, and achieved accuracy levels. Therefore, this comparative analysis is

necessary to assess the effectiveness and contribution of this study within the existing literature. As a concrete form of this analysis, the benchmarking of previous research is summarized in Table 2.

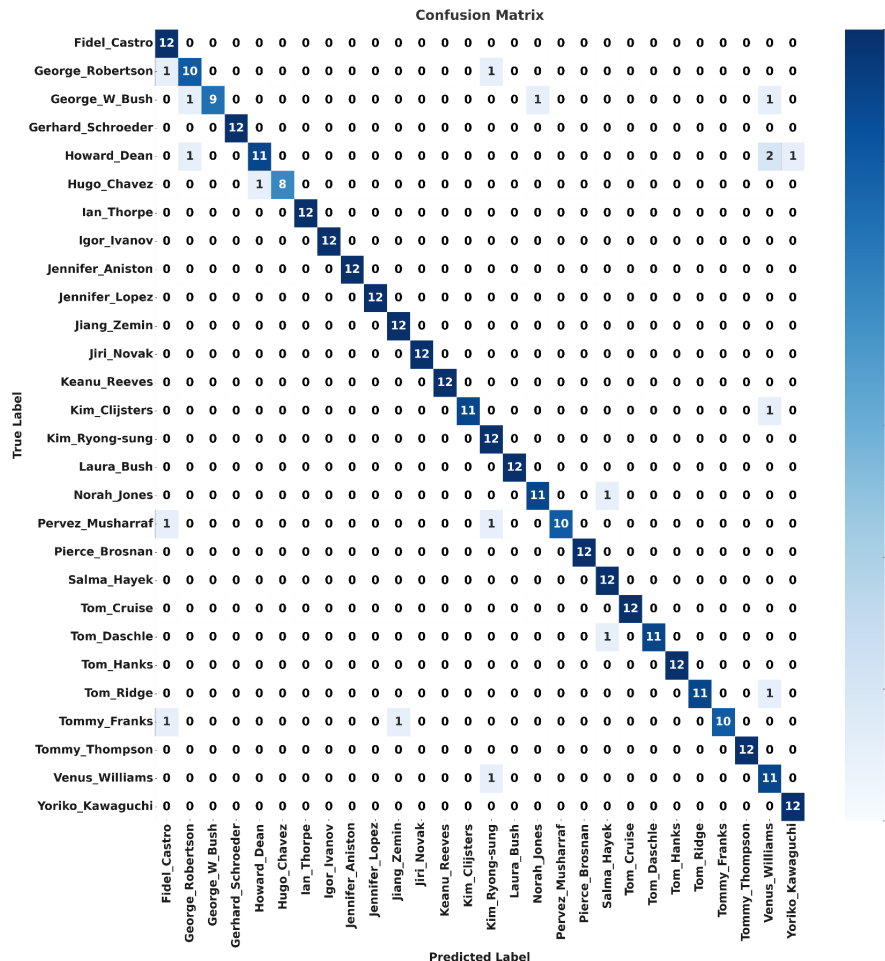


Figure 7. Confusion matrix illustrating classification results using the 80:20 data split, with strong diagonal dominance indicating correct predictions across 28 identity classes

Table 2 includes the methods, types of datasets, and accuracy achieved, and compares them with the results of this study using a combination.

Table 2 presents the comparison results of various studies on face detection using different methods. Lidya et al. applied a Transfer Learning approach with the DenseNet169 model on a large dataset of 8,929 images and achieved an accuracy of 96%. Lalitha et al. used the Haar Cascade method with 250 images per user and achieved an accuracy of 91.6%, demonstrating the effectiveness of this method in real-time systems despite limited data. Abdulatif et al. evaluated two CNN architectures, MobileNetV2 and InceptionV3, on a public dataset containing 1,000 images and achieved accuracies of 97.67% and 91%, respectively, demonstrating the superiority of MobileNetV2 in face detection. Meanwhile, Madhavi et al. used the DeepFace recognition method on 200 images extracted from a 180-second video and reported a face recognition accuracy of 84.62%.

However, it is important to note that the comparison across studies must be interpreted cautiously due to differences in experimental settings, dataset size, preprocessing pipelines, and target objectives. Moreover, their study focused on facial expression recognition rather than illumination variations, which is the primary focus of this study. Similarly, Lidya et al. worked on face mask detection rather than face recognition, and Lalitha et al. applied detection rather than classification. Abdulatif et al.'s models, while showing high accuracy, rely on more complex architectures that may not be suitable for lightweight or low-resource environments. In contrast, the present study combined DeepFace with a Multi-Layer Perceptron (MLP) classifier trained with backpropagation on a more diverse dataset (LFW with 1,680 images) and specifically evaluated performance under varying lighting conditions. These results indicate that the proposed DeepFace + MLP pipeline performs reliably in challenging illumination scenarios, offering a lightweight and effective alternative to more computationally intensive models. The benchmarking

presented in Table 2 is not intended to establish superiority but to contextualize the model's robustness within the scope of prior approaches. Future comparisons based on standardized datasets and evaluation protocols are encouraged for more rigorous validation. Unlike previous studies that used DeepFace or MLP independently, our work is the first to unify both with CLAHE-based preprocessing in a lightweight architecture designed explicitly for low-light facial recognition.

Table 2. Benchmarking Previous Research

Study	Method	Dataset	Accuracy
This Research	DeepFace + MLP	LFW (1.680 images)	95.16%
Lidya et al. [4]	Transfer Learning (DenseNet169)	8.929 images	96%
Lalitha et al [5]	Haar Cascade	250 images	91,6%
Abdulatif et al. [6]	CNN (MobileNetV2 & InceptionV3)	1000 images	97.67%, 91%
Madhavi et al. [8]	DeepFace Recognition	200 images	84.62%

4. Conclusions

This study presents a novel approach to face recognition under extreme lighting conditions by integrating DeepFace embeddings with a lightweight Multi-Layer Perceptron (MLP) classifier. The model leverages CLAHE-based contrast enhancement during preprocessing and eliminates the need for complex convolutional structures, making it suitable for real-time or embedded systems with limited computational resources. Through evaluation on the Labeled Faces in the Wild (LFW) dataset, the proposed model achieved its best performance in the 80:20 data-splitting configuration, with an accuracy, precision, and recall of 95.16%. These results surpass those of a conventional CNN baseline, especially in handling illumination variation, indicating the effectiveness of the combined embedding and lightweight classification strategy. The primary contribution of this research lies in integrating modern face embeddings with a simple yet efficient classification pipeline, enhanced through domain-specific preprocessing. Previous studies focused on low-light face recognition have not explicitly explored this combination, positioning this approach as a practical and computationally efficient solution. However, several limitations remain. The dataset was restricted to a single domain and lacked diversity regarding facial attributes such as expression, age, and background complexity. Moreover, the model was not evaluated across domains or in real-world environments, which may affect its ability to generalize beyond controlled conditions. Future work should consider extending the dataset with more diverse and representative facial samples to address these limitations. Testing the model under varying lighting scenarios in real-world settings would further validate its robustness. In addition, exploring alternative lightweight architectures and implementing stratified k-

fold cross-validation are recommended to improve statistical reliability and ensure better generalization across different data distributions.

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References

- [1] C. Ryando, R. Sigit, Setiawardhana, and B. S. B. Dewantara, "Face recognition for logging in Using Deep Learning for Liveness Detection on Healthcare Kiosks," *Int. J. Informatics Vis.*, vol. 9, no. 1, pp. 295–302, 2025, doi: 10.62527/joiv.9.1.2759.
- [2] C. H. Boe, K. W. Ng, S. C. Haw, P. Naveen, and E. A. Anaam, "An Automated Face Detection and Recognition for Class Attendance," *Int. J. Informatics Vis.*, vol. 8, no. 3, pp. 1146–1153, 2024, doi: 10.62527/joiv.8.3.2967.
- [3] S. Swapna Rani, "Enhancing Facial Recognition Accuracy in Low-Light Conditions Using Convolutional Neural Networks," *J. Electr. Syst.*, vol. 20, no. 5s, pp. 2140–2148, 2024, doi: 10.52783/jes.2559.
- [4] Lidya Fankky Oktavia Putri, Ahmad Junjung Sudrajad, and Vinna Rahmayanti Setyaning Nastiti, "Classification of Face Mask Detection Using Transfer Learning Model DenseNet169," *J. RESTI (Rekayasa Sist. dan Teknol. Informasi)*, vol. 6, no. 5, pp. 790–796, 2022, doi: 10.29207/resti.v6i5.4336.
- [5] S. Lalitha, K. P. Karthik, T. S. Gupta, M. V. Abhishek, and A. J. Sreekar, "AI-Driven Attendance Tracking with Haar Cascade and ResNet," *Procedia Comput. Sci.*, vol. 258, pp. 3544–3553, 2025, doi: 10.1016/j.procs.2025.04.610.
- [6] A. Ahmed Ali Aboluhom and I. Kandilli, "Face recognition using deep learning on Raspberry Pi," *Comput. J.*, no. September, pp. 3020–3030, 2024, doi: 10.1093/comjnl/bxae066.
- [7] B. Alane, Y. Terchi, and S. Bouguezal, "New Face Recognition System Based on DCT Pyramid and Backpropagation Neural Network," *Elektron. ir Elektrotehnika*, vol. 30, no. 1, pp. 68–76, 2024, doi: 10.5755/j02.eie.35897.
- [8] M. Kemidi, S. M. Mantrawadi, A. Mote, and D. R. Ulusu, "Facial Emotion Detection Using DeepFace," *2024 IEEE Flagsh. Int. BIT Conf. Next Gener. Appl. Green Energy Technol. BITCON 2024*, pp. 1–6, 2024, doi: 10.1109/BITCON63716.2024.10984960.
- [9] Y. Jiang, J. Zhu, L. Li, and H. Ma, "A Joint Network for Low-Light Image Enhancement Based on Retinex," *Cognit. Comput.*, pp. 3241–3259, 2024, doi: 10.1007/s12559-024-10347-4.
- [10] X. Wang, Y. C. Wu, M. Zhou, and H. Fu, "Beyond surveillance: privacy, ethics, and regulations in face recognition technology," *Front. Big Data*, vol. 7, 2024, doi: 10.3389/fdata.2024.1337465.
- [11] Z. Zhu et al., "WebFace260M: A Benchmark Unveiling the Power of Million-Scale Deep Face Recognition," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, pp. 10487–10497, 2021, doi: 10.1109/CVPR46437.2021.01035.
- [12] C. Oinar, B. M. Le, and S. S. Woo, "KappaFace: Adaptive Additive Angular Margin Loss for Deep Face Recognition," *IEEE Access*, vol. 11, no. 1, pp. 137138–137150, 2023, doi: 10.1109/ACCESS.2023.3338648.
- [13] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the gap to human-level performance in face verification," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, pp. 1701–1708, 2014, doi: 10.1109/CVPR.2014.220.
- [14] D. Kim, J. Jung, and J. Kim, "FeatherFace: Robust and

- Lightweight Face Detection via Optimal Feature Integration,” *Electron.*, vol. 14, no. 3, 2025, doi: 10.3390/electronics14030517.
- [15] S. Liu, H. Zhang, X. Li, and X. Yang, “Retinexformer+: Retinex-Based Dual-Channel Transformer for Low-Light Image Enhancement,” *Comput. Mater. Contin.*, vol. 82, no. 2, pp. 1969–1984, 2025, doi: 10.32604/cmc.2024.057662.
- [16] X. Yi, H. Xu, H. Zhang, L. Tang, and J. Ma, “Diff-Retinex: Rethinking Low-light Image Enhancement with A Generative Diffusion Model,” *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 12268–12277, 2023, doi: 10.1109/ICCV51070.2023.01130.
- [17] A. George, C. Ecabert, H. O. Shahreza, K. Kotwal, and S. Marcel, “EdgeFace: Efficient Face Recognition Model for Edge Devices,” *IEEE Trans. Biometrics, Behav. Identity Sci.*, vol. 6, no. 2, pp. 158–168, 2024, doi: 10.1109/TBIOM.2024.3352164.
- [18] M. Kim, A. K. Jain, and X. Liu, “AdaFace: Quality Adaptive Margin for Face Recognition,” *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 2022-June, pp. 18729–18738, 2022, doi: 10.1109/CVPR52688.2022.01819.
- [19] M. Azamy, A. B. Ariwibowo, and I. Mardianto, “Face Recognition Implementation with MTCNN on Attendance System Prototype at Trisakti University,” *Indones. J. Bank. Financ. Technol.*, vol. 1, no. 1, pp. 73–88, 2023.
- [20] D. Kurniadi, H. Leslie, H. Spits, and W. Suparta, “Predicting Student Performance with Multi-Level Representation in an Intelligent Academic Recommender System using Backpropagation Neural Network,” *ICIC Express Lett. Part B Appl.*, vol. 12, no. 10, pp. 883–890, 2021, doi: 10.24507/icicelb.12.10.883.
- [21] Atulanandjha, “LFW - People (Face Recognition),” vol. 6, no. 2, p. 1, 2020.
- [22] H. T. Ho, L. Vuong Nguyen, T. Le Thi Huong, and O. J. Lee, “Face Detection Using Eigenfaces: A Comprehensive Review,” *IEEE Access*, vol. 12, no. July, pp. 118406–118426, 2024, doi: 10.1109/ACCESS.2024.3435964.
- [23] Y. C. Chen, Y. S. Liao, H. Y. Shen, M. Syamsudin, and Y. C. Shen, “An Enhanced LBPH Approach to Ambient-Light-Affected Face Recognition Data in Sensor Network,” *Electron.*, vol. 12, no. 1, 2023, doi: 10.3390/electronics12010166.
- [24] I. M. Mohammed and N. A. M. Isa, “Contrast Limited Adaptive Local Histogram Equalization Method for Poor Contrast Image Enhancement,” *IEEE Access*, vol. 13, no. April, pp. 62600–62632, 2025, doi: 10.1109/ACCESS.2025.3558506.
- [25] R. C. Damale and B. V. Pathak, “System Using Machine Learning Algorithms,” *Proc. Second Int. Conf. Intell. Comput. Control Syst. (ICICCS 2018)*, no. Iciccs, pp. 414–419, 2018.
- [26] L. Lin, S. Papabathini, X. Wang, and S. Hu, “Robust Light-Weight Facial Affective Behavior Recognition with CLIP,” *arXiv Prepr. arXiv2403.09915*, pp. 607–611, 2024, doi: 10.1109/MIPR62202.2024.00103.
- [27] H. M. Shyni and E. Chitra, “PulmonU-Net: a semantic lung disease segmentation model leveraging the benefit of multiscale feature concatenation and leaky ReLU,” *Automatika*, vol. 65, no. 2, pp. 641–651, 2024, doi: 10.1080/00051144.2024.2314918.
- [28] M. Rostami, A. Farajollahi, and H. Parvin, “Deep learning-based face detection and recognition on drones,” *J. Ambient Intell. Humaniz. Comput.*, vol. 15, no. 1, pp. 373–387, 2024, doi: 10.1007/s12652-022-03897-8.
- [29] A. Meng, “Convolutional Neural Network in Face Recognition Application Analysis,” *2023 IEEE Int. Conf. Control. Electron. Comput. Technol. ICCECT 2023*, pp. 101–104, 2023, doi: 10.1109/ICCECT57938.2023.10141370.
- [30] R. Alharbey, A. Banjar, Y. Said, M. Atri, and M. Abid, “A Human Face Detector for Big Data Analysis of Pilgrim Flow Rates in Hajj and Umrah,” *Eng. Technol. Appl. Sci. Res.*, vol. 14, no. 1, pp. 12861–12868, 2024, doi: 10.48084/etasr.6668.
- [31] A. S. Al-Qazzaz, P. Salehpour, and H. S. Aghdasi, “Robust DeepFake Face Detection Leveraging Xception Model and Novel Snake Optimization Technique,” *J. Robot. Control*, vol. 5, no. 5, pp. 1444–1456, 2024, doi: 10.18196/jrc.v5i5.22473.
- [32] K. Eldora, E. Fernando, and W. Winanti, “Comparative Analysis of KNN and Decision Tree Classification Algorithms for Early Stroke Prediction: A Machine Learning Approach,” *J. Inf. Syst. Informatics*, vol. 6, no. 1, pp. 313–338, Mar. 2024, doi: 10.51519/journalisi.v6i1.664.
- [33] I. Veritawati, F. Maspiyanti, R. Mastra, E. Fernando, and A. Murtako, “Comparison of Classification Algorithms in Bamboo Distribution Mapping for Identification of Industrial Supporting Raw Materials,” *JOIV Int. J. Informatics Vis.*, vol. 9, no. 4, p. 1792, Jul. 2025, doi: 10.62527/joiv.9.4.3072.